Sec - A

Q No 1

(A)

lim $= l_1 : 2x - y + 2 = 4$

$l_2 : \quad 4x - 6y + 2z = 3$

$n_1 = 2x - y + 2 = 4 = \langle 2, -1, 2 \rangle$

$n_2 = 4x - 6y + 2z = 3 = \langle 4, -6, 2 \rangle$

$$n_1 \times n_2 = \begin{vmatrix} i & j & k \\ 2 & -1 & 1 \\ 4 & -6 & 2 \end{vmatrix}$$

$n_1 \times n_2 = i(-2+6) - j(4-4) + k(-12+4)$

$= 4i - 0j - 8k$

$= \langle 4, 0, -8 \rangle$

Scanned with CamScanner

Sec - A

Now:-

$$2x - y + 2 = 4 \quad (1)$$
$$4x - 6y + 2z = 3$$

By taking z-intercept:-

$$2x - y = 4 \quad \text{--- (i)}$$
$$4x - 6y = 3 \quad \text{(ii)}$$

Multiply eq (i) by 2

$$\cancel{4x} - 2y = 8$$
$$-\cancel{4x} \mp 6y = -3$$
$$4y = 5$$
$$\boxed{y = 5/4}$$

By putting y in equation

$$4x - \cancel{6}^{3}\left(\frac{5}{\cancel{4}_{2}}\right) = 3$$

Scanned with CamScanner

Sec-A

$$4x - 15/2 = 3$$

$$\frac{2(4x) - 15}{2} = 3$$

$$\cancel{8x - 15 = 6}$$

$$2(4x) - 15 = 3 \times 2$$

$$2(4x) = 6 + 15$$

$$4x = \frac{6+15}{2}$$

$$= \left(\frac{6+15}{2}\right) \times 4$$

$$\boxed{x = 21/8}$$

So, $\langle 21/8, 5/4, 0 \rangle$

Scanned with CamScanner

Sec-A

Parameterization:-

$$\left.\begin{aligned} x &= x_0 + v_1 t \\ y &= y_0 + v_2 t \\ z &= z_0 + v_3 t \end{aligned}\right\} \quad \begin{aligned} x &= 21/8 + 4t \\ y &= 5/4 + 0t \\ z &= 0 - 8t \end{aligned}$$

(b)

$a(t) = -6\sin t\, i - 6\cos t\, j + 2k$

$v(0) = 6i + j$

$r(0) = 6i + 6j$

we know for velocity

$\int \vec{a}(t)\, dt = v(t)$ so

$\int \vec{a}(t)\, dt = \int (-6\sin t\, i - 6\cos t\, j + 2$

Scanned with CamScanner

Sec - A

$$= -\int 6\sin t\, i - \int 6\cos t\, j + \int 2k$$

$$= -6\int \sin t\, i\, dt - 6\int \cos t\, j\, dt + 2\int k\, dt$$

$$= -6(-\cos t\, i) - 6\sin t\, j + 2tk + c$$

$$= 6\cos t\, i - 6\sin t\, j + 2tk + c$$

$$v(0) = 6i + j$$

$$v(0) = 6\cos(0)i - 6\sin(0) + 2(0)k + c$$

$$\cancel{6i} + j = \cancel{6i} + c$$

$$\boxed{c = j}$$

Scanned with CamScanner

Sec — A

$$\boxed{= 6\cos t\, i - 6\sin t\, j + 2t\, k \rightarrow j}$$

Scanned with CamScanner